

Name: _____ Date: _____

Period: _____

Problems 1–6

1. Use the unit circle to evaluate each expression. If the value is undefined, write “undefined.”

(a) $\tan(0) =$ _____ (b) $\tan\left(\frac{\pi}{4}\right) =$ _____ (c) $\tan\left(\frac{\pi}{2}\right) =$ _____

(d) $\tan(\pi) =$ _____ (e) $\tan\left(\frac{3\pi}{4}\right) =$ _____ (f) $\tan\left(-\frac{\pi}{4}\right) =$ _____

2. For each function, state the **period** and the location of **all** vertical asymptotes (write a general formula involving integer k).

(a) $g(\theta) = \tan\left(\frac{\theta}{3}\right)$

Period = _____

Asymptotes: $\theta =$ _____

(b) $h(\theta) = \tan(4\theta)$

Period = _____

Asymptotes: $\theta =$ _____

(c) $p(\theta) = -3\tan(\theta) + 2$

Period = _____

Asymptotes: $\theta =$ _____

3. A function is given by $f(\theta) = \tan\left(\theta - \frac{\pi}{6}\right)$.

(a) What is the phase shift? _____

(b) Find the vertical asymptote closest to $\theta = 0$ with $\theta > 0$. Show your work. _____

4. Describe in words how the graph of $g(\theta) = 3\tan(2\theta)$ differs from the graph of $f(\theta) = \tan(\theta)$. Address the period and vertical scale. _____

5. (AP-Style MC) Which of the following is a vertical asymptote of $g(\theta) = \tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right)$?

(A) $\theta = \frac{\pi}{4}$

(B) $\theta = \frac{\pi}{2}$

(C) $\theta = \pi$

(D) $\theta = \frac{3\pi}{2}$

(E) $\theta = 2\pi$

Answer: _____

6. The table below shows values of a function q .

θ	$-\pi/4$	0	$\pi/4$	$\pi/2$	$3\pi/4$
$q(\theta)$	-1	0	1	undef.	-1

- (a) Is q consistent with $q(\theta) = \tan \theta$? Explain. _____

- (b) What is the period of q suggested by the table? _____

Extension optional

Extension (Optional): Show algebraically that $\tan(\theta + \pi) = \tan \theta$ for all θ in the domain. Start from $\tan(\theta + \pi) = \frac{\sin(\theta + \pi)}{\cos(\theta + \pi)}$ and use the angle addition identities $\sin(\theta + \pi) = -\sin \theta$ and $\cos(\theta + \pi) = -\cos \theta$. —

Preview — Lesson 3.9: Inverse Trigonometric Functions

In Lesson 3.9 we will ask: if we know a sine, cosine, or tangent value, can we find the angle? To answer that question we restrict the domain of each trigonometric function to make it invertible, then define arcsin, arccos, and arctan. The asymptote behavior you studied today — $\tan \theta \rightarrow \pm\infty$ as $\theta \rightarrow \pm\pi/2$ — is precisely what tells us the range of arctan.